The Survey Of Atmospheric Neutrinos

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APHY 471: Introduction to Neutrino Astronomy
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Spring 2026

Abstract

Atmospheric neutrinos are produced when high-energy cosmic rays collide with particles, particularly in Earth's atmosphere. This creates showers of particles. Of the particles that are created, the most important are pions and kaons. From these collisions, the pions will decay into muons and muon neutrinos. Then these produced muons create more neutrinos. The kaons will also decay into neutrinos, which will mostly consist of muon neutrinos and a small fraction of electron neutrinos. Neutrinos are of interest because they can deepen our understanding of cosmic rays and the fundamental properties of neutrinos. In this paper, we discuss a broad overview of atmospheric neutrinos, with experiments such as Super-Kamiokande and IceCube.

1 Introduction & Background

Neutrinos are the most abundant particles in the universe with mass. The neutrino particle was introduced in the 1930s by theoretical physicist Wolfgang Pauli. He proposed this method to conserve energy, momentum, and spin in nuclear beta decays [4]. Since then, neutrinos have been the focus of various subfields of physics, including particle physics and astrophysics. This is due to neutrino oscillations and flavor-changing processes beyond the Standard Model.

A well known source of neutrino would be atmospheric neutrinos. These neutrinos are produced when high-energy cosmic rays, to be more precise, the nuclei in the cosmic ray, collide with nuclei in Earth's atmosphere. These collisions produce particle showers, commonly referred to as "cascades" [1], which visually look like

$$p + A_{\text{atm}} \longrightarrow \pi^{\pm}, K^{\pm}, K^{0}, ... + X. \tag{1}$$

Of the particles that are created from the cascades, the most important are the charged pions and kaons. These mesons will decay into muons and neutrinos, and then the muons will decay even further to create even more neutrinos. Atmospheric neutrino fluxes are dominated by muon neutrinos, with a small fraction from electron neutrinos.

Atmospheric neutrinos are mainly useful for studying neutrino oscillations [2]. This is because atmospheric neutrinos are produced naturally in Earth's atmosphere and then travel long distances before interacting with a detector. This is not the case for all atmospheric neutrinos; some can be produced near a detector, whereas others may be produced on the opposite side of the Earth before reaching the detector and being observed. On top of this, atmospheric neutrinos come in a wide range of energies.

These two factors enable experimental physicists to study neutrino oscillations and determine their probabilities. For example, the probability of a neutrino to oscillate from a to oscillate into b is [3]

$$\mathbb{P}_{a\to b} = \sin^2(2\theta)\sin^2\left(\frac{1.27\Delta m^2(\text{eV})^2 L(\text{km})}{E_\nu(\text{GeV})}\right). \tag{2}$$

Atmospheric neutrinos are most valuable because of their wide range of energies and distances. The direction of the atmospheric neutrino toward the detector determines how far it traveled. Since the oscillation probability depends on the distance and neutrino energy, this can create new and different ideas in oscillation physics. By comparing theoretical predictions with experimental values, models can be developed to test mixing parameters.

Most notably, the Super-Kamiokande experiment in Japan presented strong evidence for neutrino oscillations between flavors at a conference in 1998 [3]. This discovery proved that neutrinos have nonzero masses and changed how physicists view neutrinos. Furthermore, atmospheric neutrinos have been foundational for studying mixing parameters, the neutrino mass ordering problem, and many other open questions in neutrino physics. Despite neutrinos interacting only weakly, large underground and under-ice detectors can observe secondary charged particles produced by neutrino interactions, which emit Cherenkov light. This paper will examine the experiments and Cherenkov detectors conducted by Super-K and IceCube to advance understanding of atmospheric neutrinos, neutrino flavor oscillations, and future directions in neutrino research.

2 Production of Atmospheric Neutrinos

Atmospheric neutrino production begins with cosmic rays. Cosmic rays are high-energy particles that are mostly made up of protons and other atomic nuclei. They travel long distances through space and then enter Earth's atmosphere. Upon entering Earth's atmosphere, the nuclei in the cosmic rays collide with nuclei in the atmosphere, mostly oxygen and nitrogen. These collisions then produce the cascades of secondary particles. This process is commonly referred to as "atmospheric showers". The most important particles produced from these showers are the charged pions and kaons. These particles decay quickly and are observed as

$$\pi^+ \longrightarrow \mu^+ + \nu_\mu \tag{3}$$

This then decays into

$$\mu^+ \longrightarrow e^+ + \nu_e + \bar{\nu}_\mu \tag{4}$$

Similar decay chains occur for negatively charged particles, which produce antineutrinos. As a result, atmospheric showers produce both electron and muon neutrinos. At lower energies,

this decay chain indicates that the ratio of muon neutrinos produced is twice that of electron neutrinos. This is because the muon neutrinos are produced in both the pion and muon decays.

This predicted flavor ratio is important because it provides experimental physicists with a baseline for neutrino experiments. So, experiments should be able to observe this predicted amount. However, this was not the case; instead, the experimenters observed fewer muon neutrinos than expected. This would come to be known as the atmospheric neutrino anomaly. This anomaly called into question the accuracy of the detectors and the prediction model.

3 Neutrino Flavors and Oscillations

There are three flavors of neutrinos: electron, muon, and tau. The naming convention is derived from the charged particle it is associated with. For instance, a charged pion decaying in the atmosphere can produce a muon neutrino. When the neutrino is created, it has a specific flavor. This flavor can change as the neutrino travels distances. This is what an oscillation is. Note that a neutrino's flavor is not the same thing as its mass.

The difference lies in how the oscillation operates. In quantum mechanics, a neutrino with a specific flavor can be a combination of different mass eigenstates. These are defined by the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix [4] :

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{13} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{5}$$

As neutrinos propagate, their masses evolve differently; hence, the mixture changes over time. For example, a neutrino that was detected as a muon neutrino could be detected as a tau neutrino later. The key point is that the neutrino does not disappear. The quantum state evolves as it propagates. In the case of atmospheric neutrinos, the most important oscillation is:

$$\nu_\mu \longrightarrow \nu_\tau \tag{6}$$

This shows that the experimenters observed fewer muon neutrinos than previously expected. They instead observed muon neutrinos oscillating into tau neutrinos. Tau neutrinos are hard to detect with atmospheric neutrino experiments. This is important because neutrino oscillation can only happen if the neutrino has a nonzero mass difference. Atmospheric neutrino oscillations showed that the massless neutrino in the original Standard Model was incomplete.

4 Super-Kamiokande

Super-Kamiokande, also known as Super-K, is one of the most important neutrino detectors in the history of particle physics. To be more specific, Super-K's 1998 atmospheric neutrino paper holds one of the most remarkable discoveries, the neutrino oscillation. Super-K's experiment analyzed a 33.0 kiloton-year exposure and discovered that atmospheric neutrinos exhibited a zenith-angle-dependent shortage of muon neutrinos.[3] In other words,

the number of detected muon neutrinos depended on the direction from which they arrived. Upward-moving neutrinos that traveled through the Earth before detection were observed in less than the predicted amount. This result did not agree with the prediction of no oscillation. However, it was consistent with the muon neutrinos oscillating into the tau neutrinos [3].

The importance of this result is more than evident. Before Super-K's published 1998 result, other experiments saw small snippets of something happening with atmospheric neutrinos. Super-K provided a clear direction to what was going on with the atmospheric neutrinos. This detector distinguished between muon and electron events. This allowed physicists to observe that the event deficit affected only muon neutrinos, not all atmospheric neutrinos. Further proof of the oscillation explanation.

Super-K was able to make these observations because of its detector design. The detector has 50,000 tons of ultrapure water and has over 11,000 photomultiplier tubes. The idea is that a neutrino interacts with water and can produce a charged particle, such as a muon or an electron. If the charged particle moves faster through the water than the speed of light through the water, it is said to emit Cherenkov radiation. This radiation will form a ring of light that can be picked up by the surrounding photomultiplier tubes. In general, muon events produce sharper rings than electron events. This is because electrons scatter more frequently, producing electromagnetic showers. This is how Super-K determined whether an event was caused by an electron or muon neutrino.

This result was influential in the development of the Standard Model. This stated that neutrinos were massless. This was not the case because neutrino oscillations required neutrinos to have non-zero mass differences. This further underscores the importance of this detector for particle physics, as Super-K has shown that the Standard Model is incomplete.

5 IceCube

IceCube is another major neutrino detector; it differs from Super-K. Instead of using a tank of ultrapure water, IceCube uses a cubic kilometer of Antarctic ice at the South Pole for the detection medium [5]. This detector comprises many optical sensors within the ice block. The sensors are called digital optical modules (DOMs), and they also detect Cherenkov radiation emitted by charged particles moving through the ice. Similarly, IceCube does not directly observe neutrinos. It observes the secondary charged particles produced when neutrinos interact with the detector's matter.

IceCube was designed to test higher-energy neutrinos from astrophysical sources. Nonetheless, atmospheric neutrinos were important to this experiment. Atmospheric neutrinos are produced constantly when cosmic rays hit Earth's atmosphere. This gave IceCube a natural abundance of neutrinos. In this case, atmospheric neutrinos served as both a source and background noise for IceCube. In other words, they are useful when physicists study oscillations, but are not useful when trying to study neutrinos from outer space.

One of the most important parts of IceCube's atmospheric neutrino research is DeepCore, which is a more dense region of the detector and is located lower in the ice. DeepCore lowers the energy threshold for IceCube, making oscillation studies more feasible [6]. This is important because, as mentioned earlier, IceCube was originally designed for higher-energy

neutrinos. However, oscillations are more easily observed at lower energies. With the help of DeepCore, IceCube can measure atmospheric neutrinos that have traveled long distances, such as through the Earth, and compare their oscillation behavior with predictions.

IceCube's atmospheric neutrino measurements are important because they study neutrino oscillations at a wider energy range than Super-K did; Super-K focused on lower energies, while IceCube uses ice and observes the neutrinos over a much larger volume. Despite these differences, IceCube and Super-K agree on the same results, that atmospheric muon neutrinos change flavor as they travel.

6 Future Applications

Atmospheric neutrinos still play a major role in neutrino research. Although the discovery of neutrino oscillations is well known, there remain a few open questions in particle physics. Among these questions, the most important for atmospheric neutrinos is the neutrino mass ordering problem. Many experiments show that neutrinos have different mass eigenstates, but we cannot determine their order. In other words, physicists know that the masses are not the same; we just don't know the order of them.

Atmospheric Neutrinos are a good starting point for this problem because they travel through the Earth before reaching the detector. As neutrinos interact with the matter inside the Earth, their oscillations will change based on the electron interactions inside the Earth. This is known as matter effects. This effect depends on the neutrino's energy, the distance it traveled, and whether the mass ordering is normal or inverted. So, atmospheric neutrinos are perfect for studying the neutrino mass ordering effect because physicists can study different mass orderings by observing differences in the pattern of oscillations from the neutrinos that pass through the Earth.

Currently, detectors are being designed and or upgraded to test this idea. For example, the KM3NeT/ORCA and IceCube projects. More specifically, they have been testing GeV energy ranges to see if there is a link to the mass ordering problem [7]. For ORCA, this detector is designed to be deployed in deep-sea water and will use Earth's matter to test mass ordering. Similarly, IceCube and DeepCore demonstrate that large-volume Cherenkov detectors can use atmospheric neutrinos to perform precision oscillation measurements. These experiments are superior to the original Super-K result, owing to much larger detectors, improved calibration, and larger atmospheric neutrino datasets.

Atmospheric Neutrinos are also useful when testing if the three model neutrino flavors are complete. The original model assumes that neutrinos come only as muons, electrons, and or taus. However, if physicists discover that this is not the case, then this will entail new physics. An example is the search for sterile neutrinos. Since atmospheric neutrinos span a wide range of energies, these ideas are readily testable.

Finally, another future application of atmospheric neutrinos would be neutrino astronomy. As previously mentioned, the IceCube detector uses atmospheric neutrinos as both a source and a background. Atmospheric neutrinos constitute a background for astrophysical observations from space. This means that understanding atmospheric neutrinos is important when studying events in other media, such as cosmological neutrinos and/or solar neutrinos. This is particularly important for IceCube, which is sensitive to both atmospheric and

cosmological neutrinos outside our solar system.

7 Conclusion

Atmospheric Neutrinos are produced when cosmic rays enter Earth's atmosphere and collide with nuclei, producing unstable particles that will then decay into neutrinos. These particles are usually hard to detect; however, detectors such as Super-K and IceCube have detected them and performed analyses of these charged particles to make clear inferences about atmospheric neutrinos. The most important result was the solution to the atmospheric neutrino anomaly: Super-K showed that muon neutrinos oscillated into tau neutrinos. This is majorly important because it showed that our theoretical understanding of neutrinos at the time was incomplete.

This was key evidence. Super-K was able to show that upward-going neutrinos traveled through Earth and that muon neutrinos oscillated into tau neutrinos. This was crucial, thanks to the atmospheric neutrinos, because this showed that the Standard Model was not fully correct. So atmospheric neutrinos do, in fact, hold a special place in particle physics because physicists learned so much from them over many experiments.

References

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